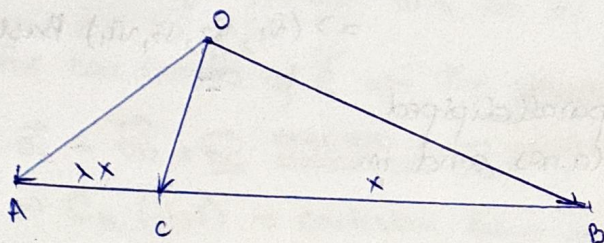


→ 10 seminars present!

(1.1.) H: $\lambda = |CA| : |CB|$, $C \in [AB]$

$$\text{cl: } \vec{OC} = \frac{1}{1+\lambda} \vec{OA} + \frac{\lambda}{1+\lambda} \vec{OB}$$

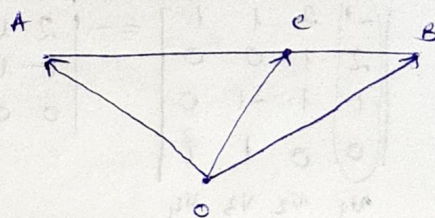


A - C - B
"collinear"

$$\lambda = |CA| : |CB|$$

$$\begin{aligned} \Rightarrow \vec{AC} &= \lambda \cdot \vec{CB} \\ (\Rightarrow) \vec{OC} - \vec{OA} &= \lambda (\vec{OB} - \vec{OC}) \\ (\Rightarrow) (\lambda + 1) \vec{OC} &= \vec{OA} + \lambda \vec{OB} \quad | : (\lambda + 1) \end{aligned}$$

$$\Rightarrow \vec{OC} = \frac{1}{1+\lambda} \vec{OA} + \frac{\lambda}{1+\lambda} \vec{OB} \quad \text{g.c.d.}$$



$$\frac{CA}{CB} = \lambda$$

$$\Rightarrow CA = \lambda CB$$

$$\vec{OC} = \vec{OA} + \vec{AC}$$

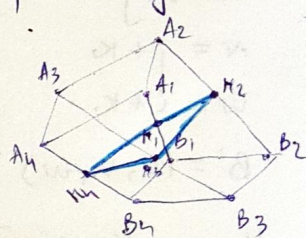
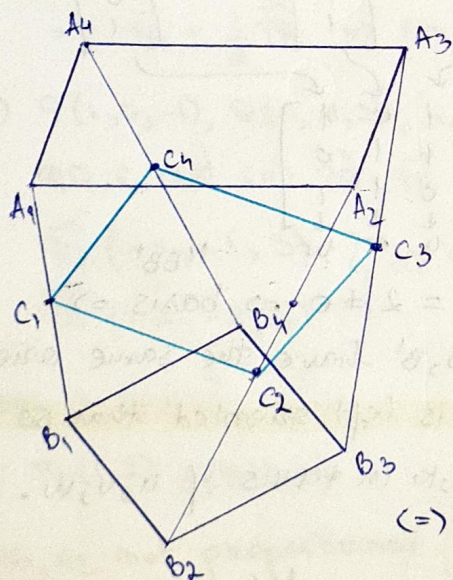
$$\vec{OC} = \vec{OB} + \vec{BC}$$

$$\Rightarrow \vec{AC} = \vec{OC} - \vec{OA}$$

$$\Rightarrow \vec{CB} = \vec{OB} - \vec{OC}$$

(1.2.) A_1, A_2, A_3, A_4 } parallelograms. $\Rightarrow \vec{A_1A_2} = \vec{A_4A_3}$ and $\vec{A_1A_4} = \vec{A_2A_3}$ (1)
 B_1, B_2, B_3, B_4 } $\Rightarrow \vec{B_1B_2} = \vec{B_4B_3}$ and $\vec{B_1B_4} = \vec{B_2B_3}$

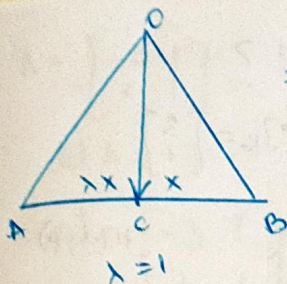
cl: midpoints of $[A_1B_1], [A_2B_2], [A_3B_3], [A_4B_4] \Rightarrow$ parallelogram.



$$\begin{aligned} C_1C_2C_3C_4 \text{ parallelogram } (\Rightarrow) & \vec{C_1C_2} = \vec{C_4C_3} \\ (\Rightarrow) & \vec{C_4C_1} = \vec{C_3C_2} \end{aligned}$$

$$\vec{C_1C_2} = \vec{C_4C_3} (*)$$

$$\Rightarrow \vec{OC_2} - \vec{OC_1} = \vec{OC_3} - \vec{OC_4}, \quad O - \text{arbitrary fixed point.}$$



$$\Rightarrow \vec{OC} = \frac{\vec{OA} + \vec{OB}}{2} \quad \text{"Median theorem in triangle"}$$

$$\vec{OC}_2 = \frac{\vec{OA}_2 + \vec{OB}_2}{2}$$

$$\vec{OC}_1 = \frac{\vec{OA}_1 + \vec{OB}_1}{2}$$

$$\vec{OC}_3 = \frac{\vec{OA}_3 + \vec{OB}_3}{2}$$

$$\vec{OB}_4 = \frac{\vec{OA}_4 + \vec{OB}_4}{2}$$

$$(*) (2) \quad \frac{\vec{OA}_2 + \vec{OB}_2}{2} - \frac{\vec{OA}_1 + \vec{OB}_1}{2} = \frac{\vec{OA}_3 + \vec{OB}_3}{2} - \frac{\vec{OA}_4 + \vec{OB}_4}{2} \quad / \cdot 2$$

$$(2) \quad \vec{OA}_2 - \vec{OA}_1 + \vec{OB}_2 - \vec{OB}_1 = \vec{OA}_3 - \vec{OA}_4 + \vec{OB}_3 - \vec{OB}_4$$

$$(2) \quad \underline{\vec{A}_1 \vec{A}_2} + \underline{\vec{B}_1 \vec{B}_2} = \underline{\vec{A}_4 \vec{A}_3} + \underline{\vec{B}_4 \vec{B}_3} \quad \text{True because (1).}$$

$\Rightarrow C_1 C_2 C_3 C_4$ parallelogram.

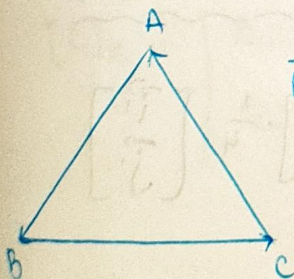
! A parallelogram can be characterised by equal vector equations (2).

1.3. u, v, w be vertices of Δ ?

a) $u(7,3), v(2,8), w(-5,5)$

b) $u(1,2,-1), v(2,-1,0), w(1,-3,1)$

$$uv + vw > uw$$



$$\vec{AB} + \vec{BC} + \vec{CA} = \vec{0}$$

$$uv \neq (-5, 5)$$

$$vw \neq$$

$$uv(-5, 5)$$

$$vw(-4, -3)$$

$$uw($$

a) $\vec{u} + \vec{w} = \vec{v} \Rightarrow \vec{u} + (-\vec{v}) + \vec{w} = \vec{0} \Rightarrow$ they form a triangle.

b) $\vec{u} + \vec{w} = \vec{v} \Rightarrow$ " — " —

OR with determinant

$$\Rightarrow 2 \text{ equal lines} \Rightarrow \det = 0. \quad \begin{vmatrix} 1 & 2 & -1 \\ 2 & -1 & 0 \\ 1 & -3 & 1 \end{vmatrix} \xrightarrow{L_3 + L_1 \rightarrow L_1} \begin{vmatrix} 2 & -1 & 0 \\ 2 & -1 & 0 \\ 1 & -3 & 1 \end{vmatrix}$$

(1.4) $K = (0, \beta)$
 $K' = (0', \beta')$

• Base change matrix from $K \rightarrow K'$?

• $[A]_K = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$, $[B]_K = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$, $[C]_K = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$\Phi = (i, j)$

$\Phi' = (i', j')$

$[0']_K = \begin{bmatrix} 6 \\ 0 \end{bmatrix} \Rightarrow 00' = 6\vec{i} + 0\vec{j}$
 (2)

$[i']_{\Phi} = \begin{bmatrix} -3 \\ 1 \end{bmatrix} \Rightarrow i' = -3\vec{i} + 1\vec{j}$

$[j']_{\Phi} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow j' = 1\vec{i} + 1\vec{j} \quad \Rightarrow \left\{ \dots \right\} \Rightarrow \begin{bmatrix} i' \\ j' \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} i \\ j \end{bmatrix}$

~~$[H]_K = \alpha\vec{i} + \beta\vec{j}$~~ $\vec{OH} = \alpha\vec{i} + \beta\vec{j}$ (1)
 $\Rightarrow [H]_K = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

$[H]_{K'} = ?$

$\vec{O'H} = \vec{O'O} + \vec{OH} = -\vec{OO'} + \vec{OH}$
 (1)
 $\stackrel{(1)}{=} -(6\vec{i} + 0\vec{j}) + \alpha\vec{i} + \beta\vec{j}$
 (2)

$= (\alpha - 6)\vec{i} + \beta\vec{j}$

$= [\alpha - 6 \quad \beta] \cdot \begin{bmatrix} i' \\ j' \end{bmatrix}$ (*)

$\begin{bmatrix} i' \\ j' \end{bmatrix} = \underbrace{\begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix}}_A \begin{bmatrix} i \\ j \end{bmatrix} \Rightarrow \begin{bmatrix} i' \\ j' \end{bmatrix} = A^{-1} \cdot \begin{bmatrix} i \\ j \end{bmatrix}$

(*) $\vec{O'H} = [\alpha - 6 \quad \beta] \cdot A^{-1} \cdot \begin{bmatrix} i' \\ j' \end{bmatrix} = [\alpha - 6 \quad \beta] \cdot \begin{bmatrix} -1 & 1 \\ 1 & 3 \end{bmatrix} \cdot \frac{1}{4} \cdot \begin{bmatrix} i' \\ j' \end{bmatrix}$

$= \frac{1}{4} \begin{bmatrix} 6 - \alpha + \beta & \alpha - 6 + 3\beta \end{bmatrix} \cdot \begin{bmatrix} i' \\ j' \end{bmatrix}$

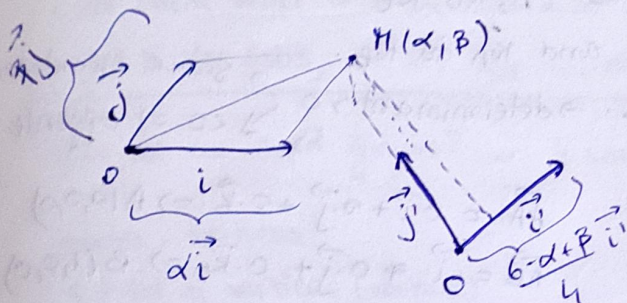
$= \frac{1}{4} \left((6 - \alpha + \beta)\vec{i}' + (\alpha - 6 + 3\beta)\vec{j}' \right)$

↳ formula for changing from one frame to another.

$$A = \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix}; A^{-1} = ? \quad A^{-1} = \frac{1}{\det A} \cdot A^* = \frac{1}{-4} \cdot \begin{bmatrix} 1 & -1 \\ -1 & -3 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -1 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\det A = -3 - 1 = -4$$

$$A^t = \begin{bmatrix} -3 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow \begin{matrix} a_{11}^* = 1 & a_{21}^* = 1 \\ a_{12}^* = 1 & a_{22}^* = -3 \end{matrix}$$



$$[H_{KK'}] = A^{-1}$$

the matrix of change of basis from ... to ... ?

→ always true or only here?

$$[H]_K = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \Rightarrow [H]_{K'} = \begin{bmatrix} \frac{6-\alpha+\beta}{4} \\ \frac{\alpha-6+3\beta}{4} \end{bmatrix} \rightarrow \text{use these for } A, B, C.$$

$$[A]_{K'} = \begin{bmatrix} \frac{6-0+2}{4} \\ \frac{0-6+3 \cdot 2}{4} \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$[B]_{K'} = \begin{bmatrix} \frac{6-6+4}{4} \\ \frac{6-6+3 \cdot 4}{4} \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$[C]_{K'} = \begin{bmatrix} \frac{6-3+1}{4} \\ \frac{3-6+3}{4} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$